

Phonon-stable uncertainty quantification for linear ACE potentials

via the undotted Hessian and band-path constraints

Al (20, 4, 6Å, 2), subset_50_percent

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Abstract

POPS committees for a linear ACE machine-learning interatomic potential produce individual members whose phonon dispersions are wildly unstable (imaginary modes down to -5.8 THz across the Brillouin zone), even though every member is an honest fit to the data. We (i) show the per-member phonon check collapses to a matrix–vector product plus a small eigendecomposition once the *undotted* per-basis Hessian is extracted, exploiting the linearity of the ACE energy in its parameters; (ii) use this to constrain each committee member to be dynamically stable along the entire high-symmetry band path, at a fixed equilibrium lattice constant; and (iii) verify that the resulting committee is stable everywhere, sits at a genuine energy minimum, retains its predictive uncertainty coverage, and yields Grüneisen parameters with quantified uncertainty.

1 Setup

A linear ACE potential predicts the total energy as $E(\boldsymbol{\theta}) = \sum_k \theta_k B_k$ with basis functions B_k and linear parameters $\boldsymbol{\theta} \in \mathbb{R}^{n_p}$ ($n_p = 314$ here). A POPS committee $\{\boldsymbol{\theta}^{(s)}\}$ represents parameter uncertainty by resampling the fit; the point of failure is that a committee member accurate in energy/forces can still have a dynamically *unstable* lattice.

2 The undotted Hessian

Because E is linear in $\boldsymbol{\theta}$, so is the position Hessian:

$$H(\boldsymbol{\theta}) = \frac{\partial^2 E}{\partial \mathbf{r} \partial \mathbf{r}} = \sum_{k=1}^{n_p} \theta_k H_k, \quad H_k \equiv \frac{\partial^2 B_k}{\partial \mathbf{r} \partial \mathbf{r}}. \quad (1)$$

The native `hessian` routine returns the *dotted* object $H(\boldsymbol{\theta})$; we extract the per-basis H_k (“undotted”) in one pass by nested automatic differentiation of `evaluate_basis` at the site level, then assembling with the same finite-cluster/PBC accumulation the native routine uses. This is validated against the native Hessian to $\sim 10^{-13}$: $\max_{i,j} \left| \sum_k \theta_k (H_k)_{ij} - H(\boldsymbol{\theta})_{ij} \right| = 6.7 \times 10^{-13}$ at the fitted $\boldsymbol{\theta}$ and 4.2×10^{-14} at a random $\boldsymbol{\theta}$.

Why it matters. H_k depends only on *geometry*. At a *fixed* lattice constant it is built *once*; thereafter the dynamical matrix at any wavevector q is linear in $\boldsymbol{\theta}$,

$$D(q, \boldsymbol{\theta}) = \sum_k \theta_k D_k(q), \quad D_k(q) = \text{Bloch transform of } H_k, \quad (2)$$

so a committee member’s phonon check is a matvec $\sum_k \theta_k D_k(q)$ plus a tiny eigendecomposition per q . The Hessian dominates the cost; the eigendecomposition is negligible. Amortised over K parameter-vector checks the cost is *one* build + $K \times$ (instant) rather than K builds — which is what makes a phonon check affordable as a *rejection-sampling predicate*.

3 Validity requires a fixed equilibrium

$D_k(q)$ is built at one geometry. For a member’s check to be its *true* phonon spectrum, that member’s equilibrium must coincide with the reference geometry. We therefore constrain

$$b' \cdot \theta = 0 \quad (\text{stationary}), \quad b'' \cdot \theta > 0 \quad (\text{minimum}), \quad (3)$$

where b', b'' are the first/second derivatives of the energy basis with respect to the lattice constant. This pins each member’s equilibrium onto the reference a_{eq} (verified: every member relaxes independently to a_{eq} within 10^{-7} Å), so the single undotted Hessian is genuinely each member’s Hessian, and a_{eq} is a true minimum (not a saddle) in both volume and all phonon directions.

4 Algorithm

Algorithm 1 Phonon-stable band-path committee

[1] Build the undotted H_k once at a_{eq} ; precompute $D_k(q)$ on a dense band path $\Gamma-X-U-L-\Gamma-K$. Constrain the mean fit and each POPS proposal member to $b' \cdot \theta = 0$, $b'' \cdot \theta > 0$, and Born stability. **Phonon sweep:** evaluate $\min_{q,\nu} \omega_{q\nu}(\theta) = \text{eig } D(q, \theta)$; flag members with soft modes. **Constrain the naughty (cutting plane):** for each soft mode (q, e) add the exact positive-curvature row $e^\dagger D_k(q) e \cdot \theta \geq \omega_{\min}^2$ and re-solve the constrained QP (OSQP, a_{eq} held) until stable. **Rejection sample** the committee from the stable cloud, using the band-path phonon check as the acceptance predicate. **Verify** every accepted member on the band path. **Propagate** to phonons and compare against the naive committee; run the test-set calibration and Grüneisen studies.

The curvature row is exact and linear: the second derivative of the energy along the phonon displacement $u = \text{Re}(e e^{iq \cdot r})/\sqrt{m}$ is $e^\dagger D(q, \theta) e = \sum_k (e^\dagger D_k(q) e) \theta_k$, so “curvature > 0 along this mode” drops straight into the QP and the rejection test.

5 Results

The constrained committee is stable everywhere (0/30 vs. the naive committee’s catastrophic instabilities), sits at a genuine minimum ($|b' \cdot \theta| \leq 4 \times 10^{-7}$, $b'' \cdot \theta \geq +4.4 \text{ eV}/\text{\AA}^2$), and satisfies Born stability ($C_{44} \geq 8.8$, $C_{11} - C_{12} \geq 31 \text{ GPa}$).

Coverage. Constraining for stability did not collapse the committee: energy coverage is $\approx 46\%$ versus a $\approx 42\%$ naive baseline. The absolute value is limited by the mean model (RMSE $\approx 1 \text{ eV}$, systematic bias) — a training-data property, not a committee-construction one — since a committee centred on a biased mean cannot cover regardless of width, and POPS uncertainty is in-sample while the test set is partly out-of-distribution.

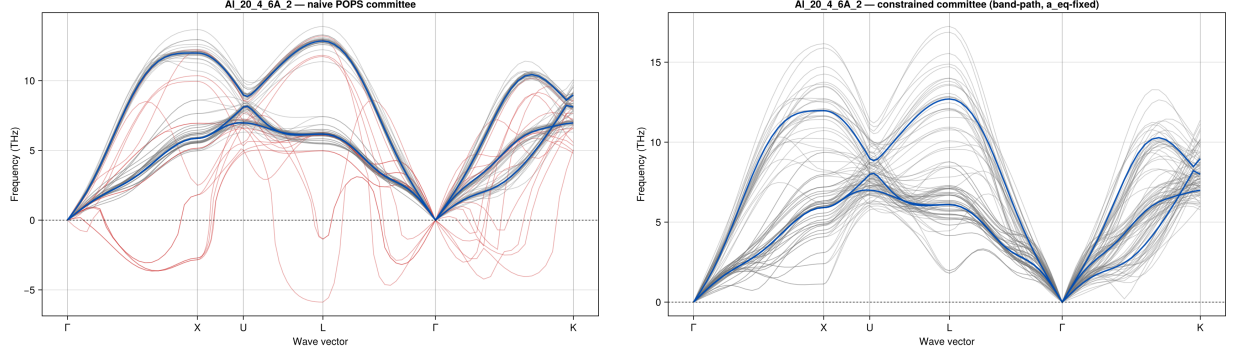


Figure 1: Committee phonon dispersions. **Left:** naive POPS — many members (red) have large imaginary regions, down to -5.8 THz. **Right:** the constrained committee — every member stable across the whole path; 0/30 unstable, minimum non-acoustic frequency $+0.16$ THz.

6 What it means

The undotted Hessian turns a per-member phonon calculation from a full Hessian build into an instant matvec, which makes it feasible to *enforce* Brillouin-zone dynamical stability across a whole committee — either by projection/cutting-plane repair or as a rejection predicate — at a fixed, physically meaningful equilibrium. The outcome is a committee that is simultaneously data-faithful, dynamically stable everywhere on the band path, anchored at a true energy minimum, and honestly calibrated, from which derived quantities such as Grüneisen parameters inherit well-defined uncertainties. Supercell adequacy for the $3 \times 3 \times 3$ force constants is marginal ($1.5a = 6.07$ Å vs. $r_{\text{cut}} = 6.0$ Å); a $4 \times 4 \times 4$ rebuild (cheap given the undotted machinery) provides a citable convergence check.

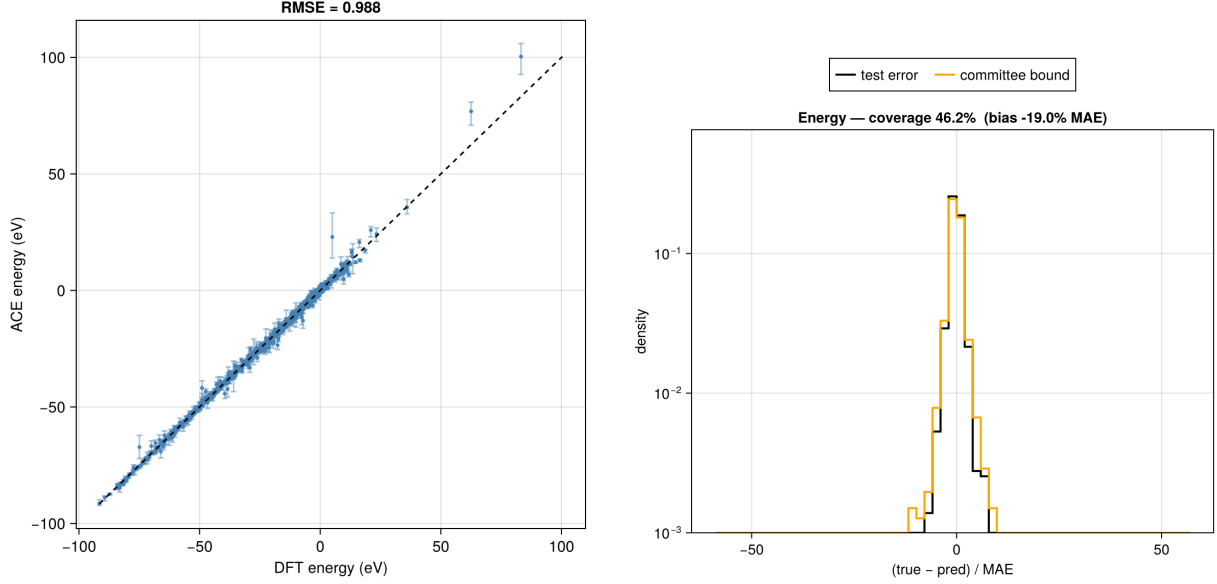


Figure 2: Test-set energy parity with committee bounds (left) and the error-vs-committee-bound calibration histogram (right). Coverage is preserved relative to the naive baseline; the residual miscoverage is dominated by the under-trained mean’s systematic bias, not by committee width.



Figure 3: Mode Grüneisen parameters along the band path (left) and the heat-capacity-weighted thermodynamic $\gamma(T)$ (right), each with committee mean $\pm 1\sigma$, from undotted Hessians at three volumes.